

Worksheet 2 – Node and Loop methods

Attempt all problems

Mention all node/loop variables on the circuit diagrams

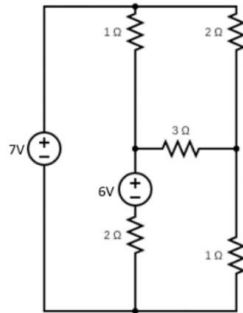
Mention all methods/equations used for each problem

Mention all assumptions (if any) made while solving

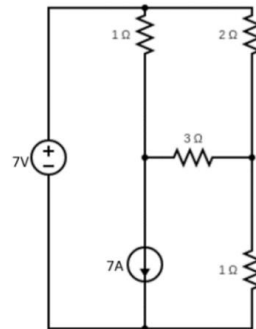
Highlight any circuit transformations used

Q1 . Solve for loop currents

(a)

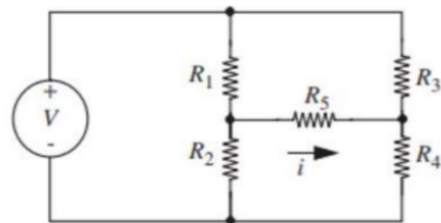


(b)



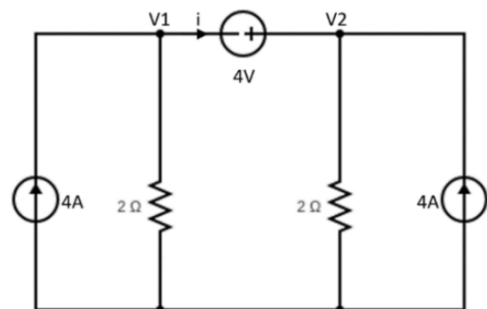
(c) In circuit (b), add a resistance of 2 ohms in series with the 7A element. Solve for loops currents. Draw your conclusion after a comparison between the two circuits.

Q2. Find all the node voltages and the current i . Given, $V=2V$, $R_3=3\Omega$, $R_1=2\Omega$, $R_4=2\Omega$, $R_2=4\Omega$, $R_5=1\Omega$

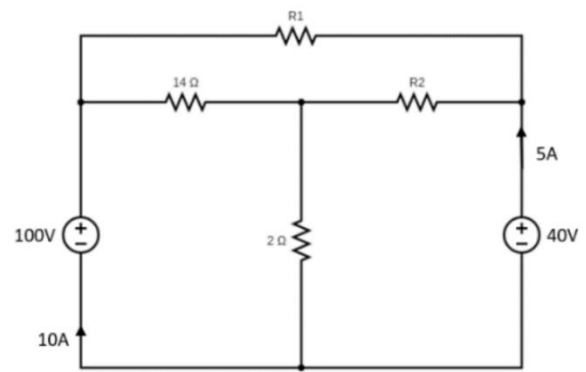


Q3. (a) For the circuit shown, find V_1 , V_2 and current i .

(b) In the same diagram, a branch consisting of a 1Ω resistor is added in parallel to the branch with the nodes V_1 and V_2 . Find the V_1 , V_2 and i now. Note your observations.

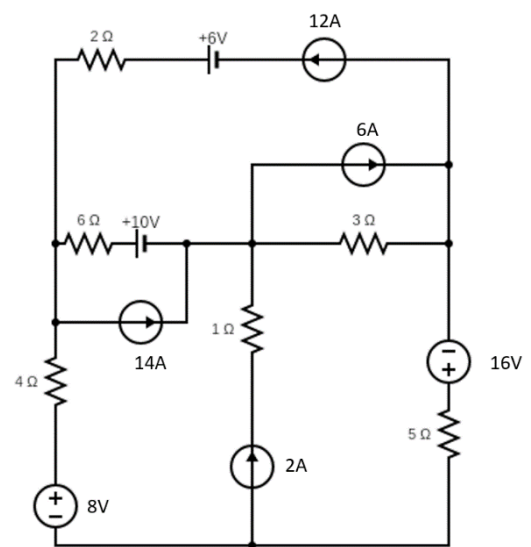


Q4. Solve for the unknown variables

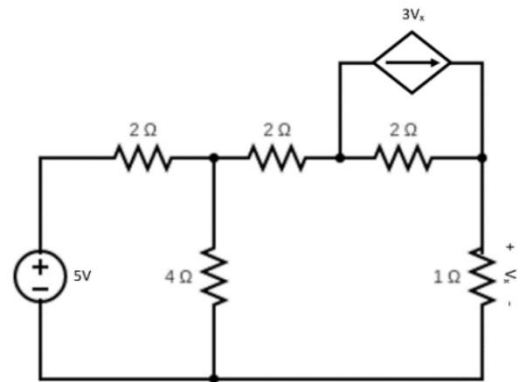


Q 5 (a) Solve for nodal voltages
 (b) In the same circuit diagram, the resistor in the 10V branch is removed. the nodal voltages now

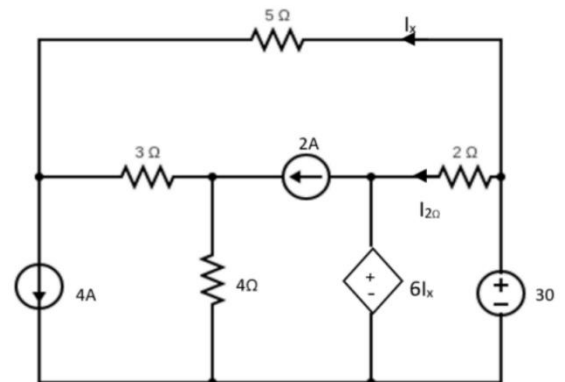
Find



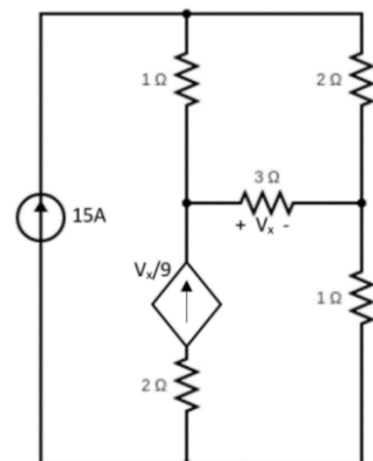
Q. 6 Find V_x



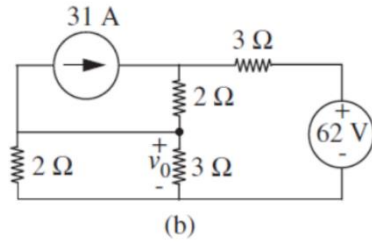
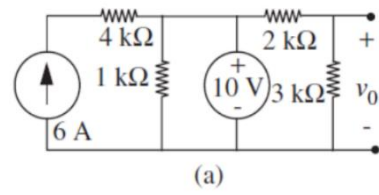
Q. 7 Solve for currents I_x and $I_{2\Omega}$ with appropriate equations.



Q. 8 Determine loop currents



Q. 9 Find voltage v_o using superposition theorem



Q. 10 Find the voltage v using superposition theorem

